

# BENXIANG XIAO

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## EDUCATION

**Southern University of Science and Technology**, Shenzhen, Guangdong, China 2023 – Present

- B.S. in Data Science (Junior Year)
- **Cumulative GPA:** 3.45/4.0 (Rank: 58/74); **Major GPA:** 3.56/4.0 (Rank: 47/74)
- **Courses:** Discrete Mathematics, Probability Theory and Mathematical Statistics, Data Structures and Algorithms, Operations Research and Optimization, Advanced Natural Language Processing, Distributed Storage and Parallel Computing
- **Research Interests:** ML Systems, Efficient LLM Inference, Distributed Systems, HPC

## PUBLICATION

[1] Jiahao Fan, Zongsheng Li, **Benxiang Xiao**, Qingzhu Zhang, Xinke Shen, and Quanying Liu. "Semantic-guided Contrastive Learning for EEG Decoding of Video Semantics." To appear in *Proceedings of the 2025 China Automation Congress (CAC)*. IEEE, 2025. (Paper ID: CCHI2025-121657)

## SKILLS

- Programming Skills: Python, C/C++, Java, Shell, CUDA, SIMD, OpenMP
- Frameworks: PyTorch, TensorFlow, PyTorch Lightning, MindSpore
- Profilers: Kunpeng DevKit, VTune, PyTorch Profiler, IPM, HipProf, MindSpore Profiler, Nsight Systems
- Languages: English, Chinese (native)

## RESEARCH & ENGINEERING EXPERIENCE

**Efficient Inference for Large Models** | Research Intern Oct. 2025 – Present

*With the group of Prof. Tianwei Zhang, Nanyang Technological University (NTU)*

- Investigating inference-efficient methods for large models, including token compression for video large models and parallel decoding strategies for diffusion language models.
- Investigated systems-oriented approaches for accelerating dynamic inference workloads while preserving model quality.

**Distributed Training Theory** | Research Intern Jun. 2025 – Feb. 2026

*With the group of Prof. Zhi Wang, Tsinghua Shenzhen International Graduate School (Tsinghua SIGS)*

- Studied theoretical aspects of efficient distributed training in cloud-edge-end architectures, including derivations for optimization behavior and convergence-related analysis.
- Explored the rationality of Adam convergence.

**High-Frequency Financial Data Distributed Computing** | Developer Sep. 2025 – Dec. 2025

*Course Project, Ranked Top 1% in Performance Benchmark*

- Built a high-concurrency quantitative factor calculation system for limit order book data on **HDFS + MapReduce**, optimizing key compression, shuffle, object reuse, and custom CSV parsing to reduce network I/O and GC overhead.
- Achieved a **13×** **speedup** over baseline with near-C/C++-level efficiency in course evaluation while preserving numerical precision. [Distributed-Storage-and-Parallel-Computing]

**2025 APAC HPC-AI Competition** | Core Member Jun. 2025 – Nov. 2025

- Optimized **NWChem** through compilation refactoring and communication tuning; profiled **DeepSeek-R1 (671B)** on multi-node GPU clusters with **Nsight Systems** and **SGLang Profiler** to identify scaling bottlenecks.

- Achieved a **4.82× speedup** for NWChem on a single node and improved DeepSeek-R1 throughput by about **70%** through TP/DP/PP tuning and backend benchmarking. [APAC\_HPC-AI]

### MindSpore Model Development Competition | Team Leader Oct. 2025 – Nov. 2025

- Profiled **DeepSeek-MoE-16B** and **Qwen1.5-MoE** using **MindSpore Profiler**; optimized expert routing, sparse GEMM, and memory fragmentation bottlenecks, and implemented a fused **Softmax+TopK** operator.
- Designed a dual-path inference engine for **Prefill** and **Decode**, reducing decoding latency by about **80%** and improving end-to-end MoE inference throughput. [MindSpore]

### 2nd SUSTech SuperComputing Competition | Team Leader Jul. 2025 – Aug. 2025

- Implemented a high-performance **HGEMM** kernel using **WMMA** on Tensor Cores, with size-specific tiling, **double buffering**, **asynchronous caching**, register swizzling, and loop unrolling.
- Achieved performance competitive with or exceeding **cuBLAS** in selected small-to-medium matrix settings while maintaining numerical accuracy. [HGEMM]

### 2025 ASC Student Supercomputer Challenge | Core Member Jan. 2025 – Feb. 2025

- Profiled the baseline workflow for m5C site detection, identified `hisat-3n-table` as the major bottleneck, and optimized synchronization, threading strategy, and MPI-based pipeline execution.
- Reduced overall runtime by **7.0×** while preserving the accuracy and reliability of the detection workflow.

### 9th Huawei ICT Competition Chinese National Finals | Team Leader Feb. 2025 – Mar. 2025

- Profiled the **Hypre** library with **Kunpeng DevKit**, optimized sparse solver kernels with **SVE intrinsics**, and improved **PCG** convergence behavior within the **BoomerAMG** framework.
- Improved computational efficiency by about **30%** for large-scale sparse matrix solving on the Kunpeng platform.

## SELECTED ACADEMIC ENGAGEMENTS

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### SupercomputingAsia (SCA) Mar. 2025 & Jan. 2026

Participated in the Student Cluster Competition Workshop at SCA, which broadened my exposure to HPC systems, large-scale computing, and performance engineering.

### Huawei Kunpeng Technical Exchange Activities Dec. 2024

Attended Huawei Kunpeng technical exchange activities related to ARM-based HPC software optimization and performance tuning.

## LEADERSHIP & COMMUNITY

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### SuperComputing Club Leader, SUSTech Jun. 2025 – Present

Organized training and project collaboration for students interested in AI systems and HPC, covering CUDA, profiling, and competition-oriented system optimization.

## HONORS AND AWARDS

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|------------------------------------------------------------------------------|-----------|
| <b>3rd Place (Onsite), 8th Place (Overall), IndySCC25</b>                    | Nov. 2025 |
| <b>Achievement Prize, ISC25 Student Cluster Competition</b>                  | Jun. 2025 |
| <b>Second Class Prize, 2025 ASC Student Supercomputer Challenge</b>          | Feb. 2025 |
| <b>Second Class Prize; Excellent AI Performance, 2025 APAC HPC-AI</b>        | Nov. 2025 |
| <b>Third Class Prize, 9th Huawei ICT Competition Chinese National Finals</b> | Mar. 2025 |
| <b>Achievement Prize, MindSpore Model Development Competition</b>            | Nov. 2025 |
| <b>Bronze Prize, MindSpore Model Fine-tuning Track</b>                       | Aug. 2024 |
| <b>Bronze Prize, 2024 Kunpeng Application Innovation Competition</b>         | Dec. 2024 |
| <b>First Prize, SUSTech SuperComputing Competition</b>                       | Aug. 2025 |
| <b>Campus Academic Star, Southern University of Science and Technology</b>   | Sep. 2024 |
| <b>Outstanding Student, Southern University of Science and Technology</b>    | Sep. 2025 |